

Ideas Parked — Not Yet Merged

Staging document; each entry marks its eventual destination

This document holds ideas discussed against Pattern Arithmetic v15.19, Inference, and Assembly, none of which have been written into any of those three files yet. Each entry states its status and its intended destination. Nothing here is settled; the point of this file is to let that be true without leaving Paper 1 mid-edit.

1 Destined for Pattern Arithmetic (untouched, remains open)

Mix across differing directions. No progress made. The existing paragraph already states the gap precisely: Mix is defined only among states sharing a direction $\hat{n}$, and what it should mean across differing directions is unsettled, with a provisional vector-sum recipe already tried and withdrawn. A concrete motivating case surfaced in discussion: a single witnessed episode carrying humidity, pressure, and air-speed readings, each naturally sitting at its own direction on the sphere, needing to be read together. This sharpens the existing open item with a real example; it does not resolve it. [**Open, unchanged.**]

2 Destined for Pattern Arithmetic (advances an existing open item)

Location as a sort — a third candidate. The existing paragraph names two candidates and adopts neither. A third was tested here: dedicate separate, disjoint dimensions to content, to time, and to location, rather than letting any of them share an axis.

[**Verified.**] Two shifts acting through disjoint content dimensions commute exactly (checked numerically, not approximately: time-then-location and location-then-time landed on the identical point to full floating-point precision). Two shifts forced to share a single content dimension do not commute, confirming the disjointness is what matters, not rotation itself.

[**Verified.**] Content stays recoverable after both shifts: projecting back onto the content dimensions alone reproduced the original direction closely, while the time-axis and location-axis each carried a clean, separately-readable amount of shift, with nothing blended between them.

[**Open, unchanged.**] The encoding itself — what specific rotation a given time value or location corresponds to — is not stated anywhere. The geometry now supports it cleanly; the mapping does not exist yet.

This is real, checkable content, not yet written as an addition to the existing paragraph.

3 Destined for Pattern Arithmetic (open item stays open, unchanged)

Reducing a path by merging close actors. The existing paragraph asks for three things: a stated distance, a merge rule, and a bound on how far a reduced path's endpoint may drift from

the unreduced one. Discussion here supplied a distance (see below) and partially addressed the merge rule, but only for magnitude, not for direction — and direction is the harder half this item is actually about.

[**Verified.**] Switching the strength calculation from the companion paper’s clip to Inference’s already-established Hill form, $r' = |z|/(K + |z|)$, changes nothing about the resulting direction $\chi' = \arg(z)$. Checked directly on two constructed sequences with identical content and opposite order: both gave the same angle under either formula. The Hill substitution fixes evidence-count blindness in strength; it does not touch the order-blindness this open item is actually asking about.

[**Open, unchanged.**] No merge rule for direction exists. No drift bound exists for either magnitude or direction. This item remains exactly as open as it was before this discussion, now with the magnitude half more precisely separated from the direction half.

4 Destined for Inference (new mechanism)

Candlestick summary: a richer merge rule for reference-atom building. Not a resolution of the Paper 1 item above — a different question. Paper 1’s open item is about *walking*: can a shortened path still be walked to roughly the same endpoint. This is about *reading*: how to summarize a group of atoms for comparison against a reference atom, which is Inference’s own established business (its reference-atom section), not *T*’s.

[**Verified.**] Open/close/high/low, computed the way a trading candle is, distinguishes exactly the case flat Mix could not. Two sequences, identical content, opposite order — naive Mix gave the identical angle (47.5) for both. The candle’s open and close swapped between them ($20 \rightarrow 160$ versus $160 \rightarrow 20$), correctly reading one as ending badly and the other as ending well. High and low stayed identical for both, correctly capturing that only the *range* was shared, not the *direction of travel*.

[**Settled, matches existing machinery.**] The scope rule — every atom belongs to exactly one time unit, no overlap between buckets — is not new; it is precisely the discrete-time-bucket metadata already defined on a walk’s own steps in the companion paper.

[**Open.**] This gives Inference a genuinely better reference-atom merge rule than flat Mix. It is not proposed as a fix for Paper 1’s walking-specific open item, which remains untouched above. Whether open/close/high/low itself needs a coherence check analogous to $\bar{R}_c$ has not been considered.

5 Destined for Inference (naming caution, not yet a mechanism)

“Inference Mandelbrot” is not Assembly’s Mandelbrot, and should not share its name. Assembly’s multi-zoom promotes an unordered thing (a settled camp) and is fully handled by the Hill bijection already verified there. Whatever would promote an *ordered* thing (a walked path, a movie, a time series) into one higher-level atom is a different, harder problem — it is the direction half of the path-reduction item above, restated from a different angle. [**Naming flag only.**] If this is ever built, it wants its own name; reusing “Mandelbrot” for it would be the same kind of collision already caught and corrected for “collapse,” “singularity,” and the quantum swap test.

6 Considered and found to need nothing filed

For completeness: the reference-atom and precursor-witness mechanism, the bundle-vector / uncapped V_{derived} distinction from Mix, and the $\pi = (x; w_1, \dots, w_n)$ poem trajectory were all discussed

at length and all turned out to already be fully present in Inference and Paper 1 respectively. Nothing from those threads is parked here, since there was nothing left unresolved to park.